

Emergency Blood Product Administration

Guideline Title: Emergency Blood Product Administration

Guideline Number: GUID-3203-PR025

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Revision dates: 6/8/2021	Developed by: Air Care Team
Approved by: Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** This department guideline is used to provide safe and effective administration of uncrossmatched blood and blood products to Air Care Team (ACT) patients with hemorrhagic shock from trauma or other mechanisms. This guideline is also used to recognize and effectively treat transfusion reactions.
- II. **DEFINITIONS:** When used in this department guideline, these terms have the following meanings.
 - a. Blood Products: Whole blood, O positive and O negative
 - b. IV – Intravenous
 - c. SBP – Systolic Blood Pressure
 - d. NS – Normal Saline
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:**
 - i. Personal protective equipment (PPE)
 - ii. Appropriate blood product
 - iii. Filtered blood tubing or leukocyte reduction filter
 - iv. A patent IV with Normal Saline infusing
 - v. Pressure bags may be utilized with caution for transfusing blood as long as the pressure inside the bag does not exceed 300 mmHg
 - b. **INDICATIONS:**
 - i. **Adult** Indications
 1. Blunt or penetrating trauma, hypovolemia with acute blood loss not limited to gastrointestinal hemorrhage, post-partum hemorrhage, or severe coagulopathy, signs and symptoms of hemorrhagic shock with:
 - a. SBP < 90 mmHg **OR**
 - b. Pulse rate >110 bpm **OR**

- c. Evidence of Peripheral Vasoconstriction including cool, pale skin and delayed capillary refill > 2 seconds.
- 2. Hemorrhage with documented HGB < 7 g/dL
- ii. **Pediatric Indications**
 - 1. Blunt or penetrating trauma, hypovolemia with acute blood loss, signs and symptoms of hemorrhagic shock with:
 - a. Evidence of hypoperfusion (pallor, peripheral vasoconstriction including cool, pale skin and delayed capillary refill > 2 seconds) **OR**
 - b. SBP < (70 mmHg + 2 x age) **OR**
 - c. Pulse Rate – clinical context should be taken into account when assessing pulse rate number (e.g., presence of fever, dehydration, anxiety, pain, etc.)
 - i. Neonates (less than 28 days) < 100 bpm or > 180 bpm
 - ii. Infant (1 month – 1 year) < 80 bpm or > 160 bpm
 - iii. Child (1 – 10 years) < 60 bpm or > 130 bpm
 - iv. Older Child (>10 years) < 60 bpm or > 110 bpm
 - 2. Hemorrhage with documented HGB < 7 g/dL
- c. **CONTRAINDICATIONS:**
 - i. Patients or patient's legal representatives' refusal of blood or blood products due to religious or personal reasons.
 - ii. Previous history of significant blood transfusion reaction.
- d. **DOSAGE:**
 - i. **Adult**
 - 1. Bolus entire unit of whole blood IV/IO, may give second unit of whole blood if indicated
 - ii. **Pediatric**
 - 1. Bolus **20 ml/kg** of whole blood IV/IO, may repeat if indicated
- e. **PROCEDURE:**
 - i. Consider the administration of Tranexamic Acid (TXA), see Air Care Team *GUID-3203-T016 Tranexamic Acid for Severe Hemorrhage in Trauma Patients*.
 - ii. Utilize appropriate PPE.
 - iii. Initially assess for the following:
 - 1. Vital signs.
 - 2. Skin condition: rashes, redness, etc.
 - 3. Size and integrity of IV site.
 - a. Adults: Prefer at least an 18 g for Whole Blood
 - b. Pediatrics: 22-24 g.
 - c. Breath sounds.
 - d. Fluid volume status.
 - e. History of previous transfusion reactions.

- iv. Blood products must be maintained at a controlled temperature of 1 – 6 degrees Celcius during transport and must be infused within 4 hours of removal from the thermal control environment.
- v. Select the appropriate blood product
 - 1. Type A whole blood for both males and females of all ages.
 - 2. Females:
 - a. Childbearing age or below: O negative (O-) Whole Blood if available.
 - b. O+ can be given to women of childbearing years in life-threatening situations if O- is not available: notify the receiving facility that patient received O+ and will require rho(d) immune globulin (RhoGAM).
 - c. Beyond childbearing age: O+ Whole Blood
- vi. Ideally blood should be warmed during administration using approved blood/fluid warmer device following manufacturer's instructions, see *GUID3202-PR026 Warming of Blood Products and Intravenous Fluids*.
- vii. Never transfuse blood in piggyback fashion into any IV with medications infusing. Only piggyback into NS Line.
- viii. Once the patient has been identified as meeting the criteria for emergent transfusion of unmatched blood:
 - 1. **Bedside Verification Process for Emergency Release Blood**
 - a. At the bedside the Transfusionist (ACT RN) or Independent Verifier (Credentialed ACT Paramedic) verifies the following information: patient name/alias, MR#, blood product label type (O Negative or O Positive), the blood tag unit number and expiration date.
 - b. Enter date and time on the designated line on the blood tag if all information matches
 - c. After verification by both the Transfusionist (ACT RN) and the Independent Verifier (Credentialed ACT Paramedic); the blood is transfused.
 - 2. May place blood in pressure bag and inflate the bag not to exceed 300 mmHg.
 - 3. Bolus entire unit Whole Blood for adult.
 - 4. See part d. ii, 1 – above - Pediatric Dosage for pediatric dosing and rate of transfusion.
 - 5. Between each transfusion:
 - a. Flush IV with NS flush.
 - b. Record time of completion on Uncrossmatched Emergency Blood Release Form.
- f. **TRANSFUSION REACTION** - Reassess airway patency, hemodynamic stability and treat any transfusion reaction or complications per *ACT*

GUID-3203-M004 Anaphylaxis for adults or ACT GUID-3203-Ped002 Pediatric Allergic Reaction/Anaphylaxis for pediatrics.

- i. Immediate signs and symptoms
 1. Allergic Reaction – hives, facial swelling, laryngeal edema, bronchospasm
 2. Febrile – fever (> 1 C change from baseline), chills, headache, flushing, tachycardia
 3. Hemolytic Reaction – sudden onset of chills, fever, flushing, lower back or chest pain. Also, may present as shock, excess bleeding, hematuria
 4. Circulatory Overload – dyspnea, tachypnea, hypertension with headache
 5. Contaminated Blood – shaking fever with dry flushed skin, severe hypotension, abdominal pain, vomiting, diarrhea.
- ii. If a transfusion reaction occurs:
 1. Stop transfusion immediately.
 2. Assess vital signs.
 3. Remove blood bag and tubing and set aside.
 4. Provide systemic support as required.
 5. Assure bag and tubing are given to receiving blood bank or RN/appropriate medical staff, upon arrival at receiving facility, along with a copy of the Uncrossmatched Emergency Blood Release Form with transfusion reaction section completed.
- iii. Once transfusion of any products is complete:
 1. Complete the Uncrossmatched Emergency Blood Release Form in its entirety (leave blood bank slip attached to units but do not complete).
 2. Notify receiving physician of any transfusion reactions and treatment provided.
 3. Notify receiving physician if O+ blood was given to a female of child bearing age.
 4. Obtain all necessary signatures.
 5. Make 2 copies of completed Uncrossmatched Emergency Blood Release Form: *original must stay with patients chart regardless of receiving facility.*
- g. **Orlando Regional Medical Center (ORMC), Arnold Palmer Hospital or Winnie Palmer Hospital as receiving facility:**
 1. Bring completed blood product bags, copy of the Uncrossmatched Emergency Blood Release Form, and a patient label to the blood bank for testing and replacement of products.
 2. Complete the Air Medical Blood Component Disposition Log immediately upon returning to aircraft base of operations.
 3. Scan and email Component Disposition Log to

r-actbloodbankqc@orlandohealth.com and retain copy at base in Blood Product Tracking binder.

4. Upload a copy of the Uncrossmatched Emergency Blood Release Form to EMScharts patient record.

h. Other hospital as receiving facility:

1. Leave Original form with patient chart. Leave a copy of the Uncrossmatched Emergency Blood Release Form with the completed blood products bags and notify facility that the form and units must get to the blood bank for testing.
 2. Proceed to ORMC, if weather and operational considerations allow, to replace used blood products.
 3. Complete the Air Medical Blood Component Disposition Log immediately upon returning to aircraft base of operations.
 4. Scan and email Component Disposition Log to r-actbloodbankqc@orlandohealth.com and retain copy at base in Blood Product Tracking binder.
 5. Upload a copy of the Uncrossmatched Emergency Blood Release Form to EMScharts patient record.
- i. If unable to restock blood products at ORMC, contact blood supplier immediately upon return to base and request replacement units be delivered. Complete Air Medical Blood Component Disposition Log immediately upon receiving units.

C. REALTED GUIDELINES and PROCESSES:

- i. *GUID3203-G002 - General Patient Care*
- ii. *GUID3203-T016 - Tranexamic Acid*
- iii. *GUID3203-M004 - Anaphylaxis*
- iv. *GUID3203-Ped002 - Pediatric Allergic Reaction/Anaphylaxis*
- v. *GUID3203-PED009 - Pediatric Hypotension*

D. DOCUMENTATION:

- i. EMScharts Blood Product Administration on page 7
- ii. Presence or absence of transfusion reaction
- iii. Vital signs before transfusion and when completed

E. Uncrossmatched Emergency Blood Release Form (No number)

F. Air Medical Blood Component Disposition Log – Form 842.3

G. REFERENCES:

- i. Air Methods Blood Products Administration for Hemorrhagic Shock
- ii. Maher, P., et al (2017). Logistical Concerns for Prehospital Blood Product Use by Air Medical Services. *Air Medical Journal* 36:263-267